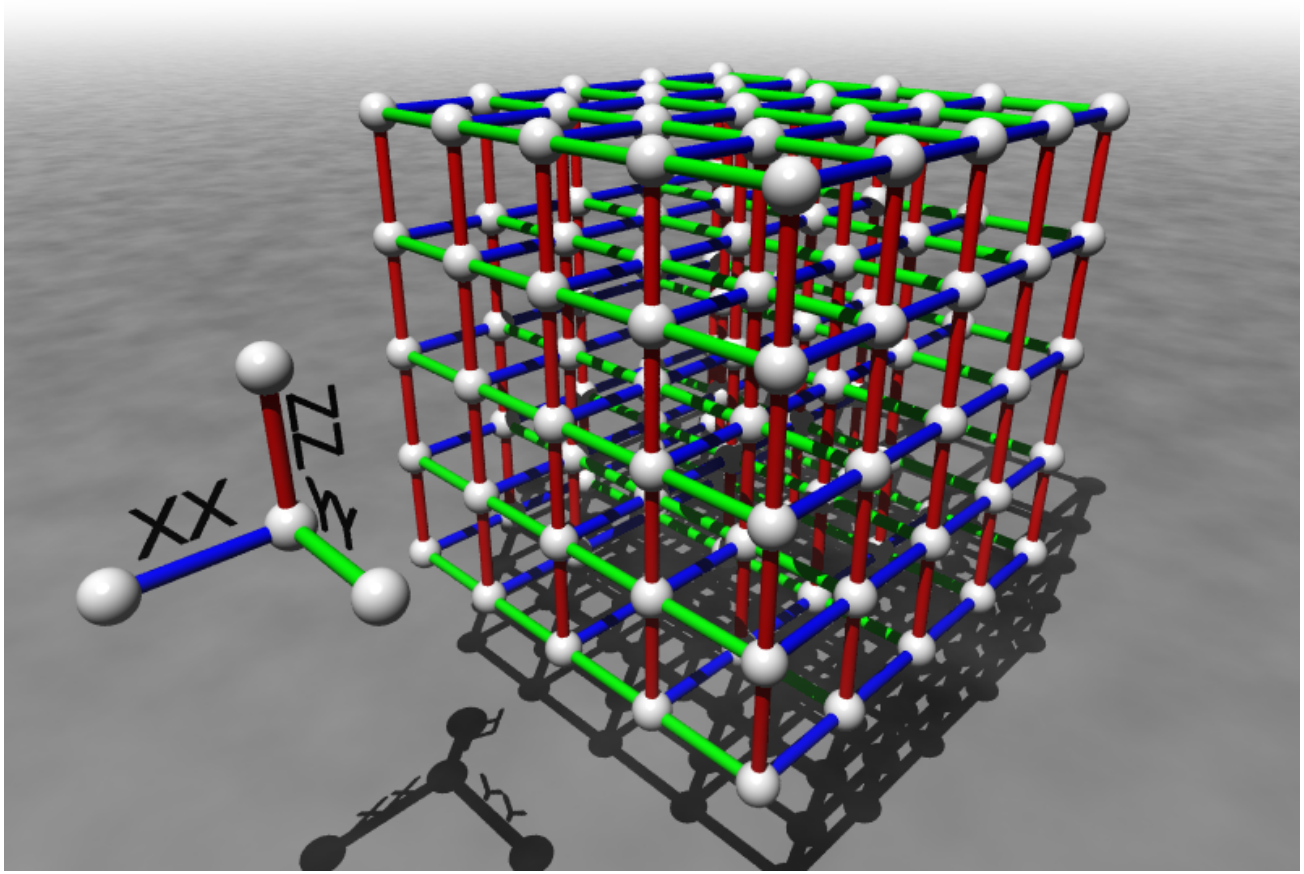
Building Robust Qubits

Dave Bacon

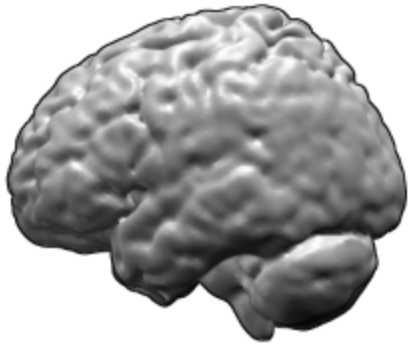


Department of Computer Science & Engineering

My Mortal Enemy



Success Stories



Life



Computers

How is it that nearly error free behavior can arise from systems whose components are unreliable?

$$|\psi\rangle \rightarrow$$

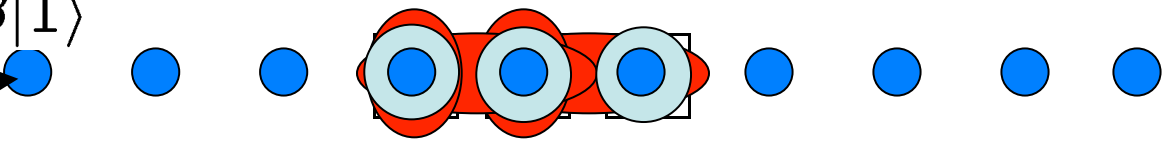


Quantum Computers

What is a Quantum Computer?

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

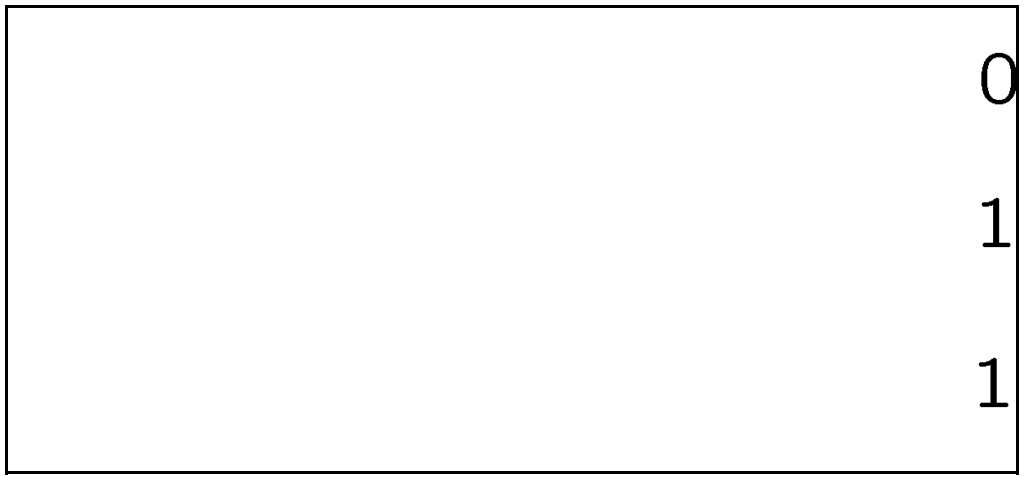
qubits



$$\frac{1}{2}(|000\rangle + |011\rangle + |100\rangle - |111\rangle)$$

Quantum algorithm

1. Initialize
2. Series of unitary gates
3. Measurement



Why Build a Quantum Computer?

Original answer: [Peter Shor 1994]

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

=

398075086424064937397125500550386491199064362342526708406385189575946388957261768583317

×

472772146107435302536223071973048224632914695302097116459852171130520711256363590397527

Best classical algorithm takes time $O(\exp(n^{1/3}))$

Quantum algorithm takes time $O(n^2 \log n)$

Breaks cryptosystems: RSA, Diffie-Hellman
(i.e. when you paid for this conference!)



"The problem of distinguishing prime numbers from composites, and of resolving composite numbers into their prime factors, is one of the most important and useful in all of arithmetic... The dignity of science seems to demand that every aid to the solution of such an elegant and celebrated problem be zealously cultivated" — Carl Gauss

Why Build a Quantum Computer?

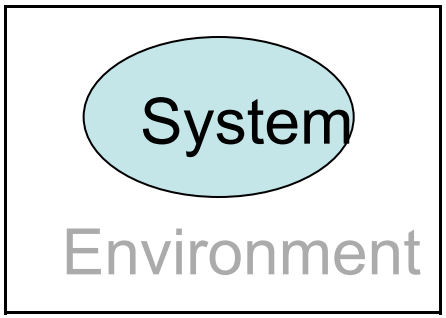
Today's answer:

- Factoring, discrete log [Shor 94]
- Unstructured search [Grover 96]
- Various hidden subgroup problems [Long List]
- Pell's equation [Hallgren 02]
- Hidden shift problems [van Dam, Hallgren, Ip 03 Childs, van Dam 05]
- Graph traversal [CCDFGS 03]
- Spatial search [AA 03, CG 03/04, AKR 04]
- Element distinctness [Ambainis 03]
- Various graph problems [DHHM 04, MSS 03,…]
- Testing matrix multiplication [Buhrman,Špalek 04]

- Simulation of quantum systems ??!!??

Obstructions

DECOHERENCE



Total Hamiltonian

$$H = H_S \otimes I + I \otimes H_E + H_I$$

system environment interaction

open quantum systems can become *entangled* with their environment

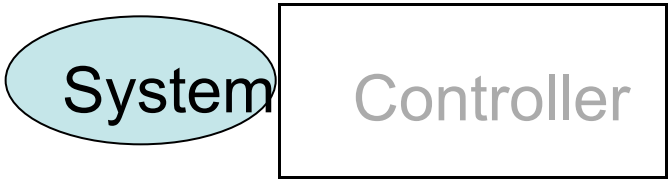
$$(\alpha|0\rangle_S + \beta|1\rangle_S) \otimes |0\rangle_E \rightarrow \alpha|0\rangle_S \otimes |0\rangle_E + \beta|1\rangle_S \otimes |1\rangle_E$$

coherent superposition incoherent sum

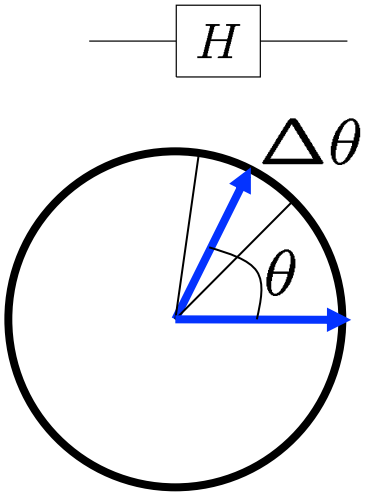
$$(\alpha|0\rangle_S + \beta|1\rangle_S) \rightarrow \rho = |\alpha|^2 |0\rangle_S\langle 0|_S + |\beta|^2 |1\rangle_S\langle 1|_S$$

More Obstructions

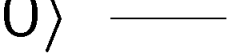
FAULTY CONTROL



Imprecise or faulty unitary gates

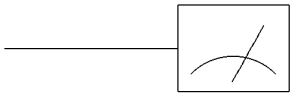


Imperfect state preparation



$$\rho \rightarrow (1-\epsilon)|0\rangle\langle 0| + \epsilon|1\rangle\langle 1|$$

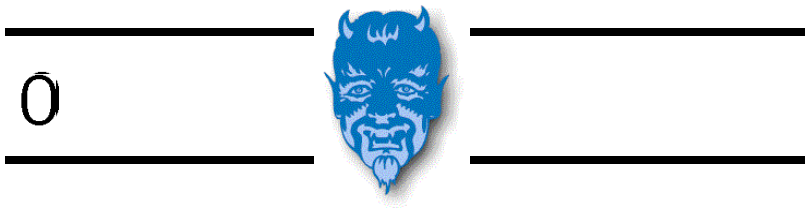
Imperfect measurements



$$\alpha|0\rangle + \beta|1\rangle \rightarrow Pr(0) = |\alpha|^2 - \epsilon$$

$$\alpha|0\rangle + \beta|1\rangle \rightarrow Pr(1) = |\beta|^2 + \epsilon$$

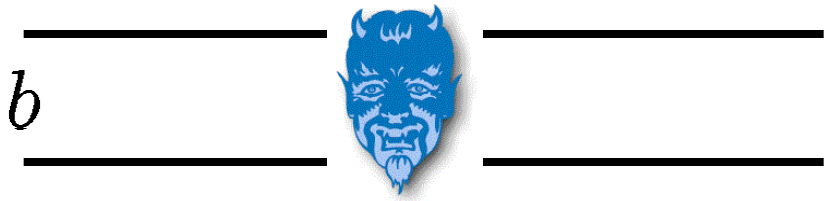
Classical Solutions



Probability of bit flip = *p*

Solution: redundancy

$0 \rightarrow 000$
$1 \rightarrow 111$



$$\left.\begin{matrix}000\\001\\010\\100\end{matrix}\right\} \rightarrow 0$$

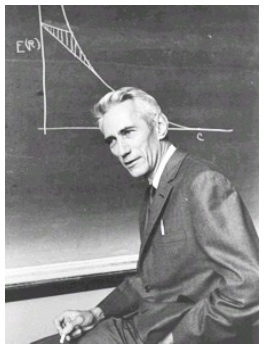
$$\left.\begin{matrix}111\\110\\101\\011\end{matrix}\right\} \rightarrow 1$$

Probability of failure =*3(1-p)p2+p3*

=3p2-2p3

<p if *p<1/2*

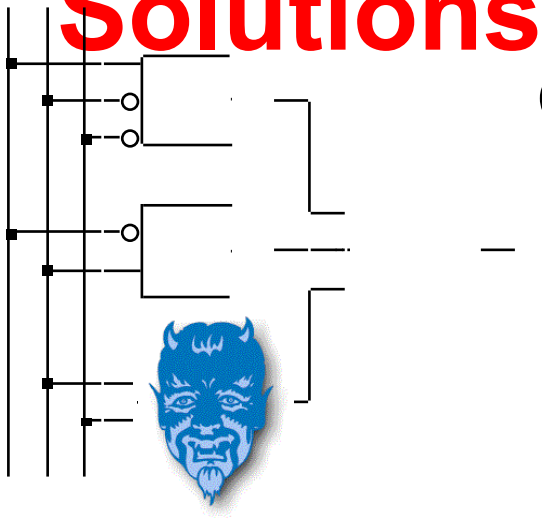
Use redundancy to protect information against noise



Claude Shannon 1948

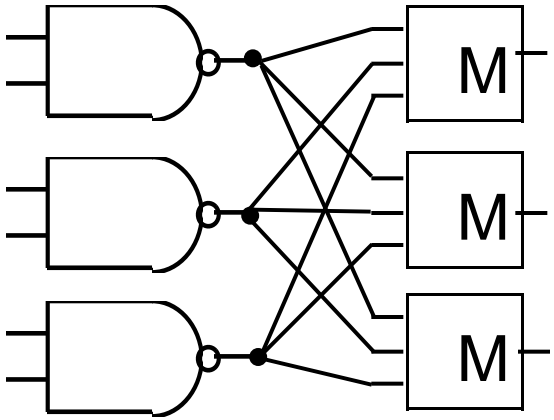
“A mathematical theory of communication”

More Classical Solutions



Classical gates fail with probability p

Solution: error correcting codes + fault tolerant construction



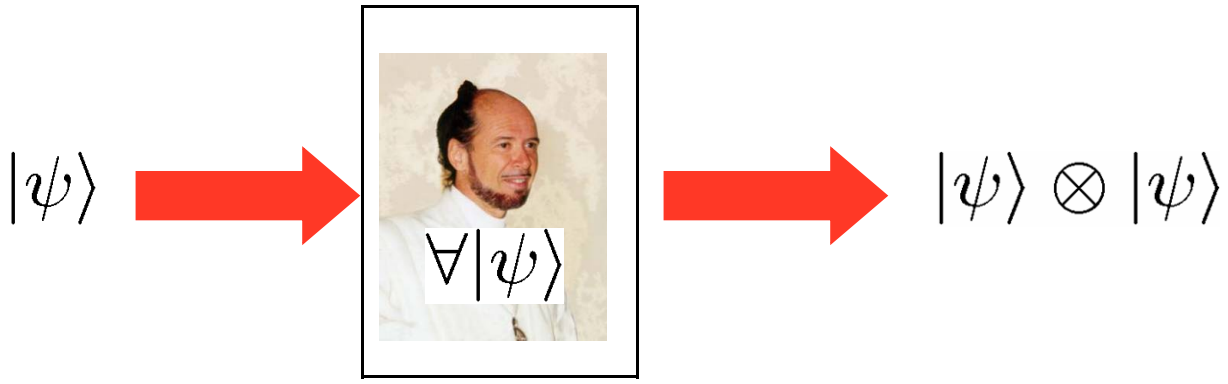


John von Neumann, 1956

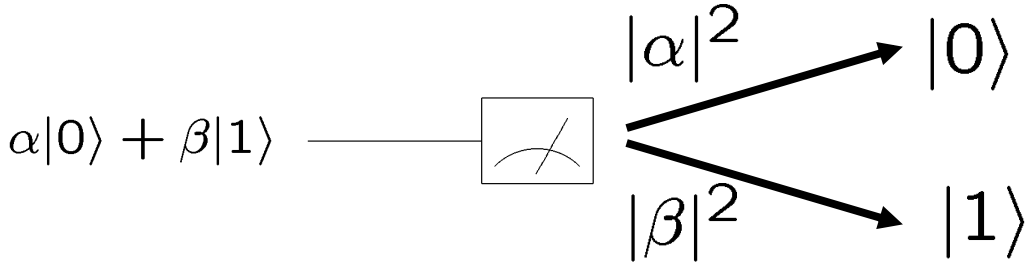
“Probabilistic logics and the synthesis of reliable organisms from unreliable components”

Quantum Problems?

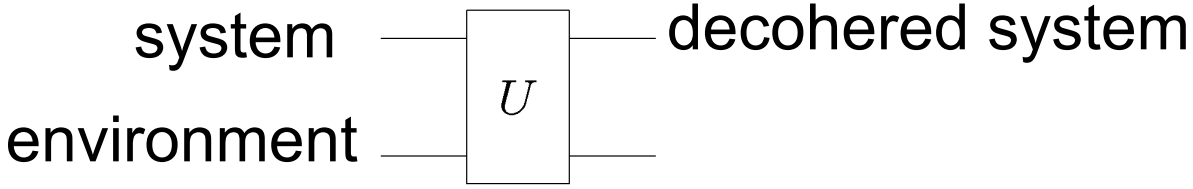
No cloning of quantum information



How to perform error correction without measuring?

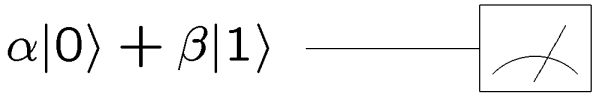


Quantum errors seem "analog"

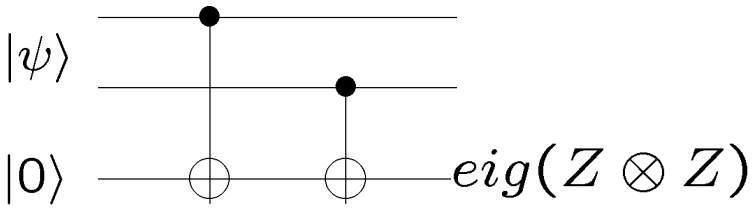


Quantum Observations

How to perform error correction without measuring?



Observation: not all measurements destroy quantum information



$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

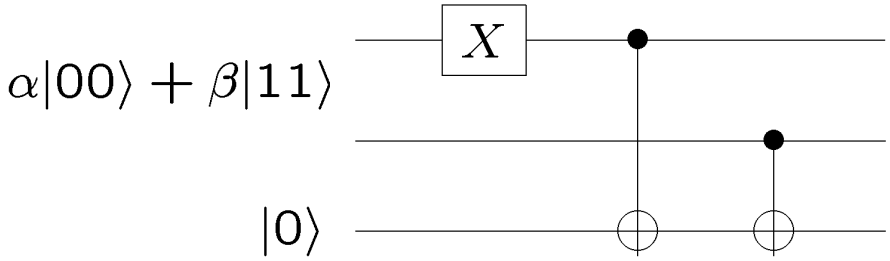
$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Information encoded into a subspace undisturbed

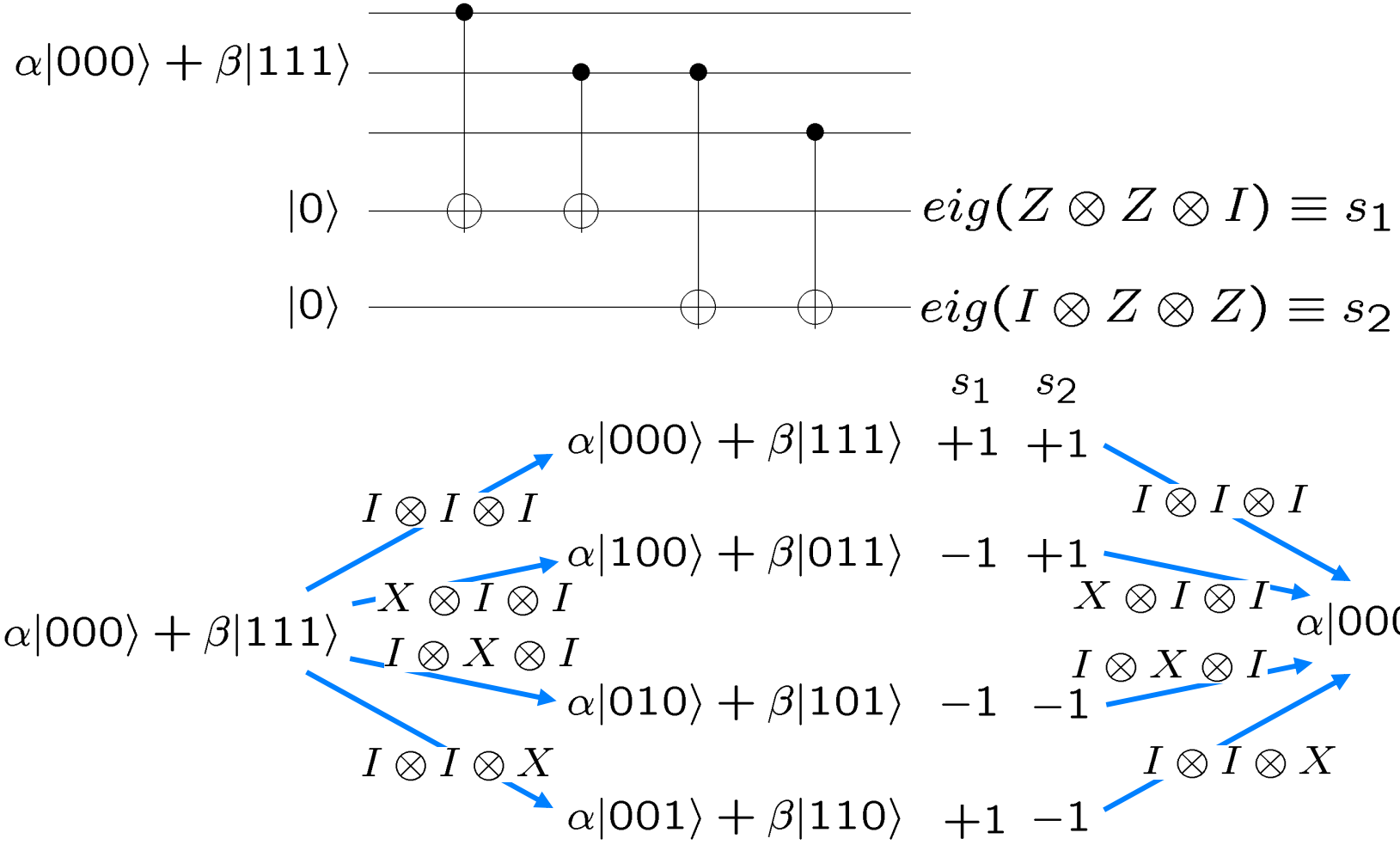
$$\alpha|00\rangle + \beta|11\rangle \rightarrow \alpha|00\rangle + \beta|11\rangle \qquad eig(Z \otimes Z) = +1$$

Can detect “leaving” of subspace



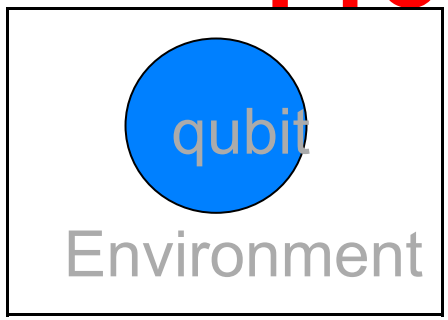
$$eig(Z \otimes Z) = -1$$

Quantum Observations

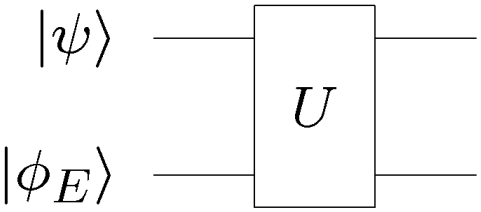


By encoding into subspaces we can correct single bit flips

Quantum Problems?



$$H = H_S \otimes I + I \otimes H_E + H_I$$



$$|\psi\rangle\otimes|E_I\rangle + (X|\psi\rangle)\otimes|E_X\rangle + (Y|\psi\rangle)\otimes|E_Y\rangle + (Z|\psi\rangle)\otimes|E_Z\rangle$$

digitized effects of decoherence

Think: amplitude for no error, *X* error, *Y* error, *Z* error

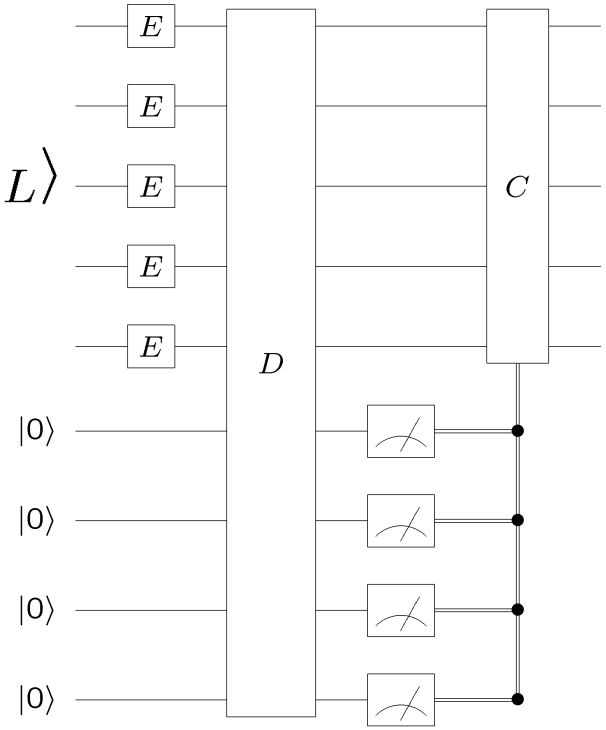
Quantum Error Correcting Codes

$$s_1 = X \otimes Z \otimes Z \otimes X \otimes I \qquad [s_i, s_j] = 0$$
$$s_2 = I \otimes X \otimes Z \otimes Z \otimes X$$
$$s_3 = X \otimes I \otimes X \otimes Z \otimes Z$$
$$s_4 = Z \otimes X \otimes I \otimes X \otimes Z$$

Encode qubit into 2 dimensional subspace with all +1 eigenvalues

Single qubit errors (i.e. $I \otimes X \otimes I \otimes I \otimes I, I \otimes I \otimes Y \otimes I \otimes I$) all lead to a unique subspace labeled by *si* eigenvalues

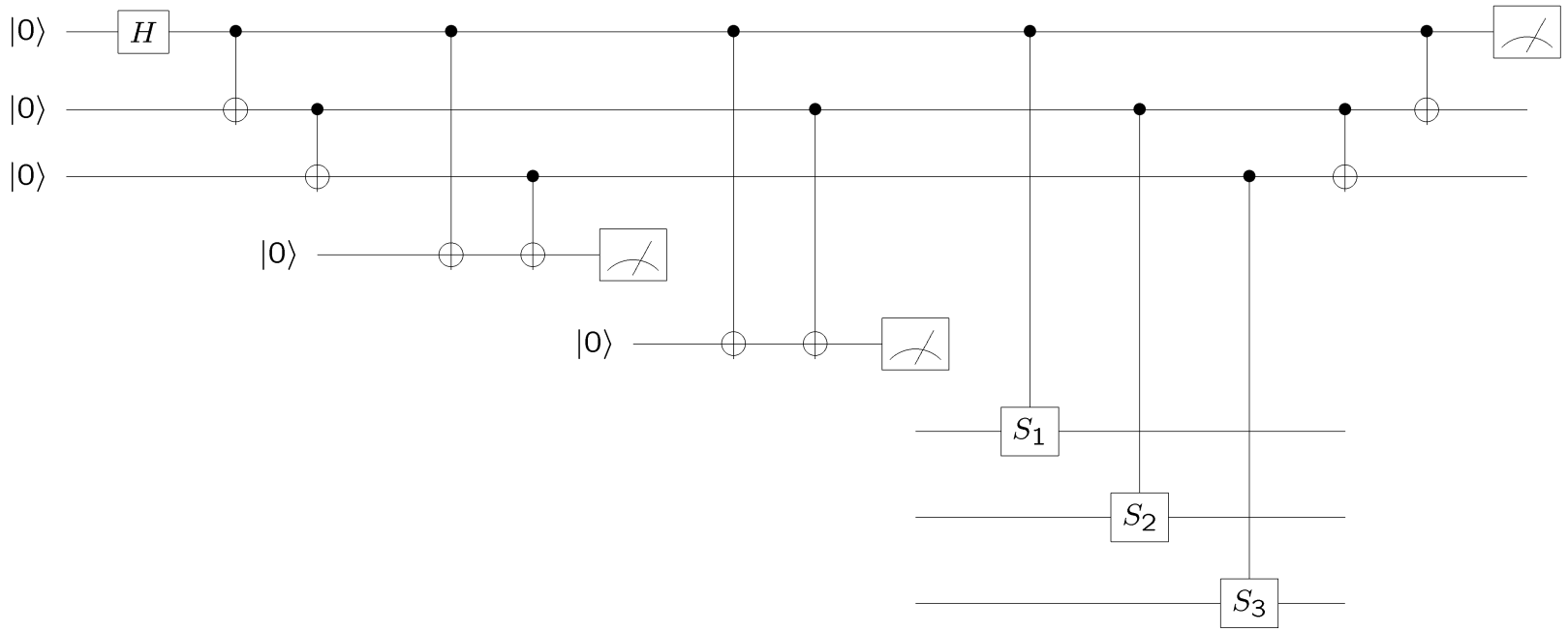
$\alpha|0_L\rangle + \beta|1_L\rangle$



$\approx \alpha|0_L\rangle + \beta|1_L\rangle$

"bare" error rate = *p*
error rate on encoded = *O(p2)*

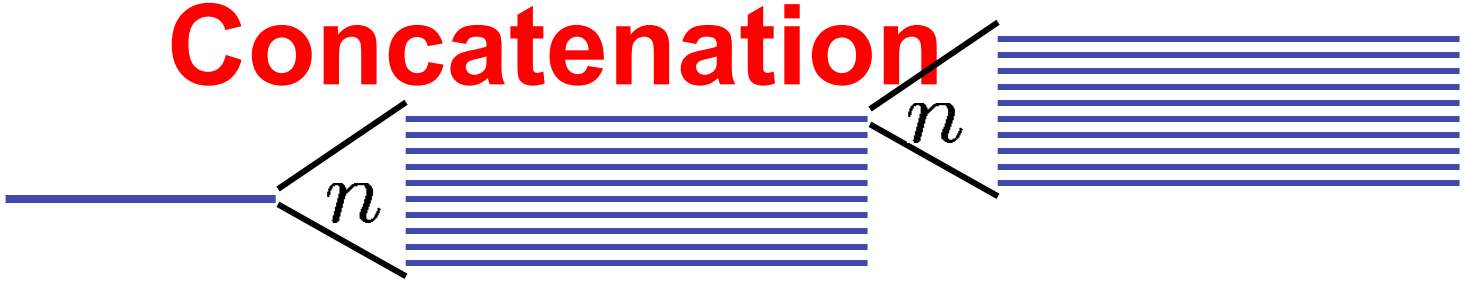
Fault-Tolerant Quantum Computation



Must design quantum circuits which succeed even when every element of the circuit (preparation, unitary gates, measurements) can fail.

Quantum error correction + Fault-tolerant procedures

Perfection Through Concatenation



failure probability

p $\quad$ $\frac{cp}{2}$ $\quad$ $\frac{c(cp2)}{2}$

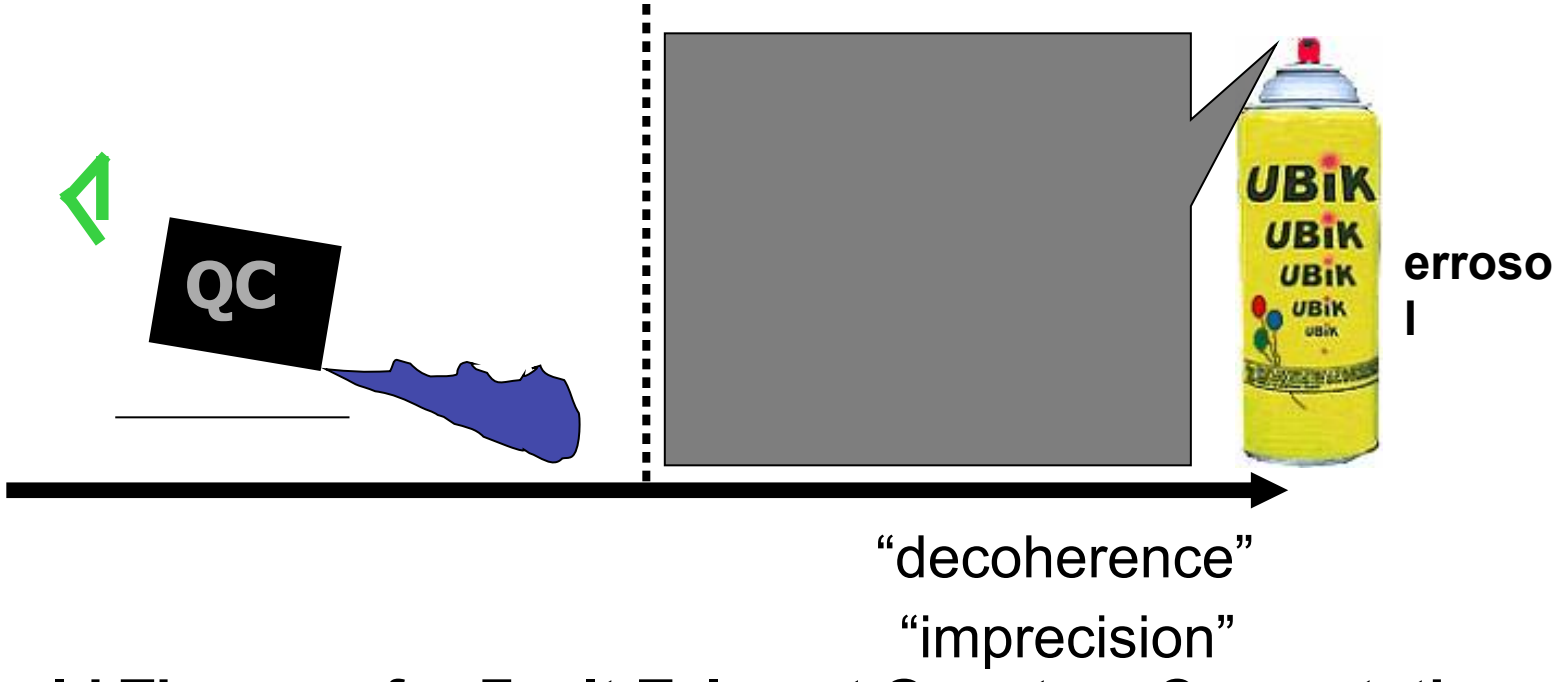
At kth level of concatenation, probability of an error which destroys the encoded quantum information is

$$p_{fail} = \frac{(cp)^{2^k}}{c} \qquad \frac{p_{fail}}{p_{th}} = \left(\frac{p}{p_{th}}\right)^{2^k} \qquad p_{th} = \frac{1}{c}$$

and uses $N_{qubits} = n^k$ qubits

$$N_{qubits} = \left[\log_{\frac{p}{p_{th}}} \left(\frac{p_{th}}{p_{fail}}\right)\right]^{\log_2 n} = O\left(\text{poly}\log \frac{1}{p_{fail}}\right)$$

Threshold Theorem, ~1996



Threshold Theorem for Fault-Tolerant Quantum Computation

[Aharonov, Ben-Or, Knill, LaFlamme, Zurek, Kitaev, Gottesman, Preskill, Shor, + others]

If noise is weak enough and control is precise enough, then a quantum computation to a desired accuracy can be carried out with an overhead polynomial in the log of the accuracy.

Threshold(s) Theorem(s)

Threshold theorems come in many different forms….

provable bounds, Monte Carlo estimates, …

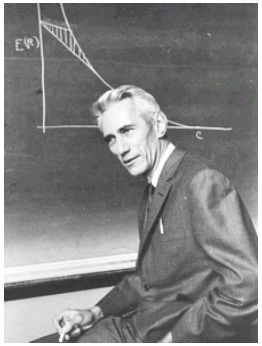
different shapes…

assumptions of error models
assumptions of physical devices
assumptions of architectures

yielding threshold values for error rates from between 10-9 to 10-2

…but device complexities are universally very high. (hundreds to thousands to millions of “bare” qubits per encoded qubit with extremely complex manipulations)

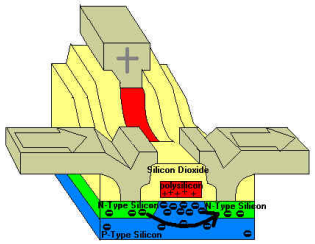
Back To Reality





theory

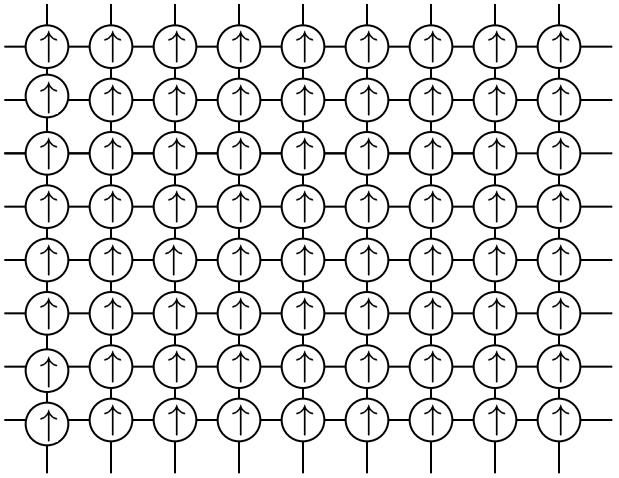




practic
e

Practical devices of information processing enact the ideas of theory in the *physics* of the devices.

Toy Model

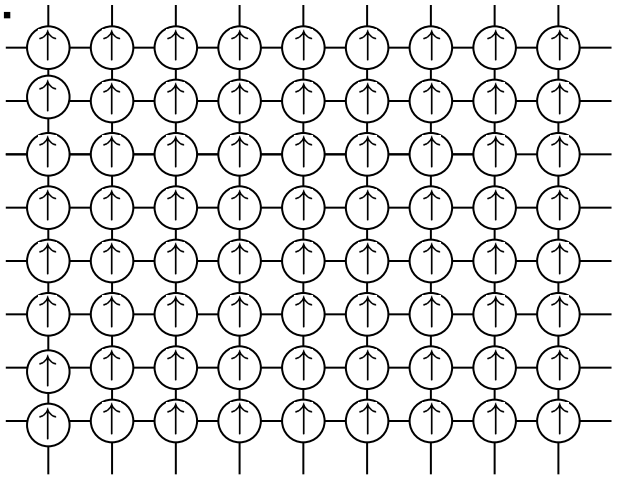


Ising model in two+ dimensions

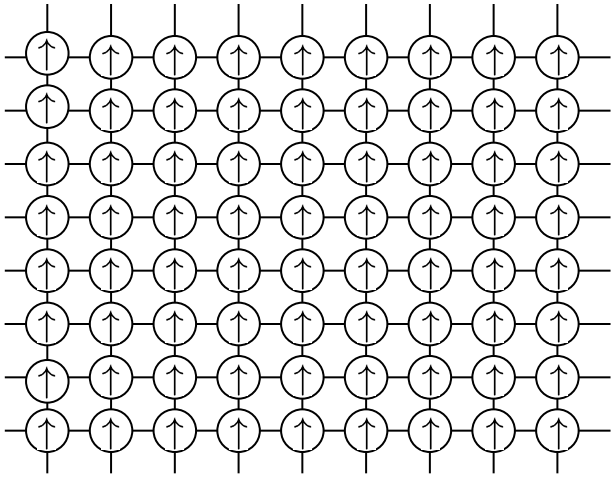
$$H = -J \sum_{\langle i,j \rangle} s_i s_j$$

Redundancy code:

$0 \rightarrow$



$1 \rightarrow$

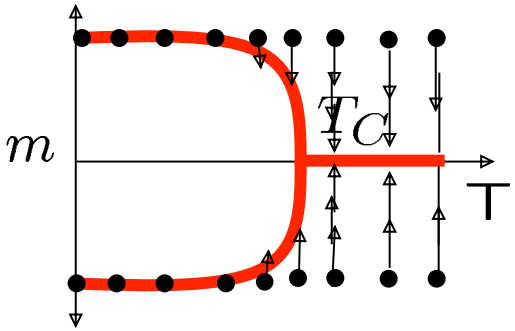




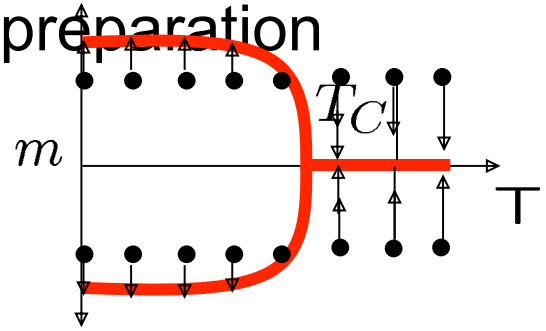
Robust



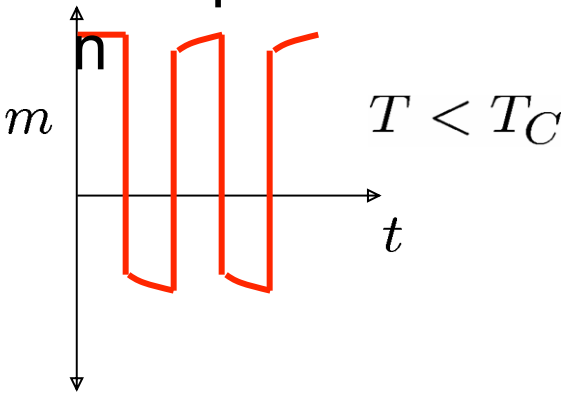
Relaxation to thermal



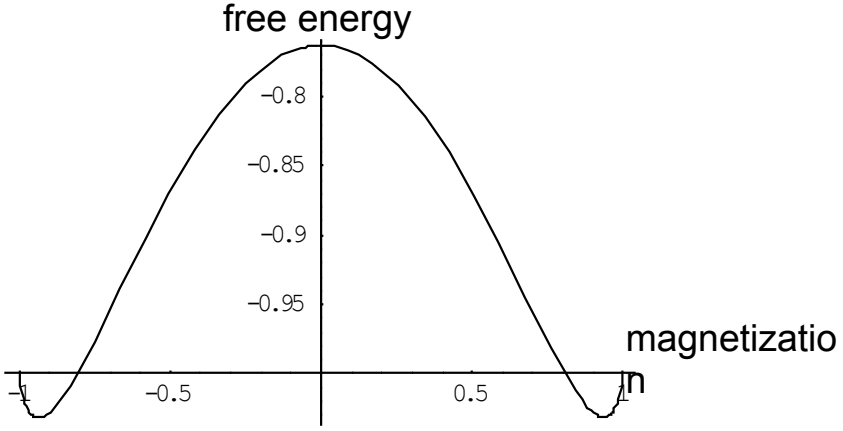
Imperfect preparation



Manipulation

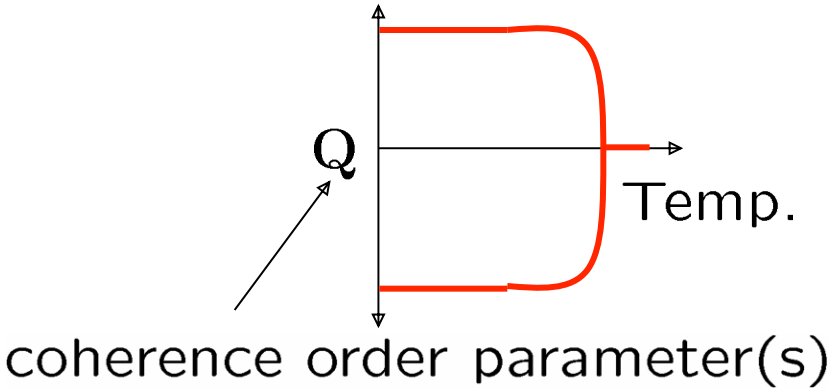


The two encoded states have lifetimes which grow with the size of the system:



Quantum Hard Drive?

Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?



Today in QC: we have theory (like von Neumann, Shannon) but not devices which implement these idea (without severe device complexity overhead.)

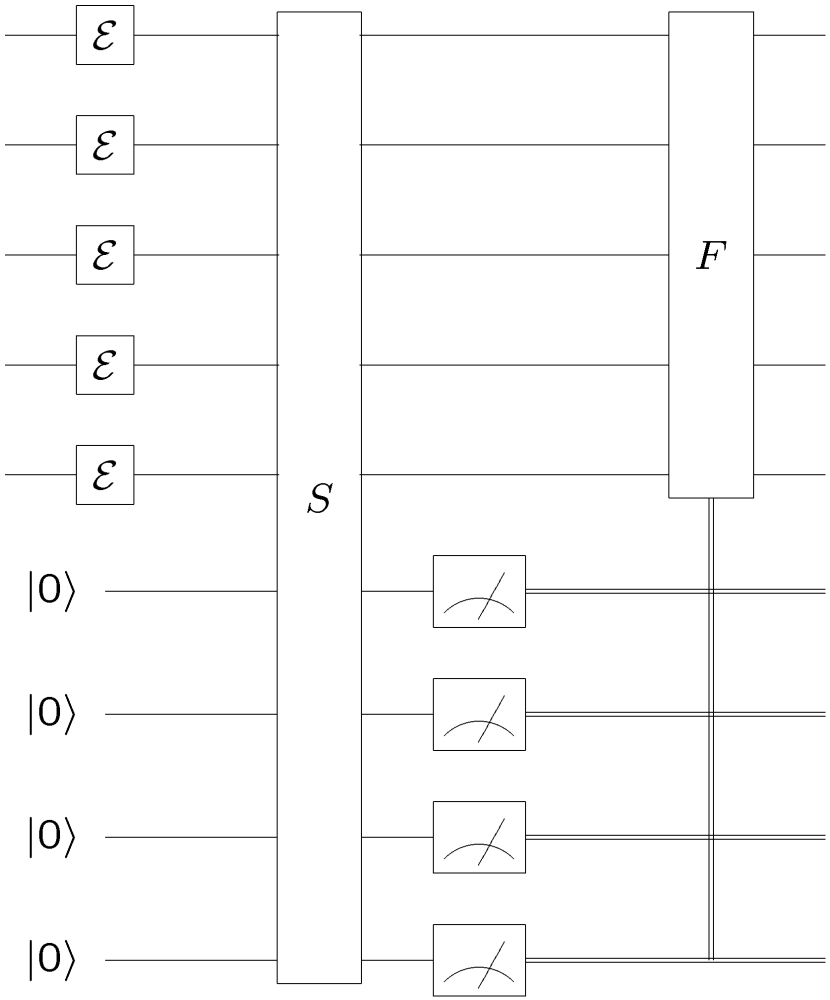
Quantum Error Correction



error

diagnose

fix



encoded quantum information

encoded quantum information

"cold" ancilla qubits

"hot" ancilla qubits

Quantum error correction is essentially (sophisticated) cooling

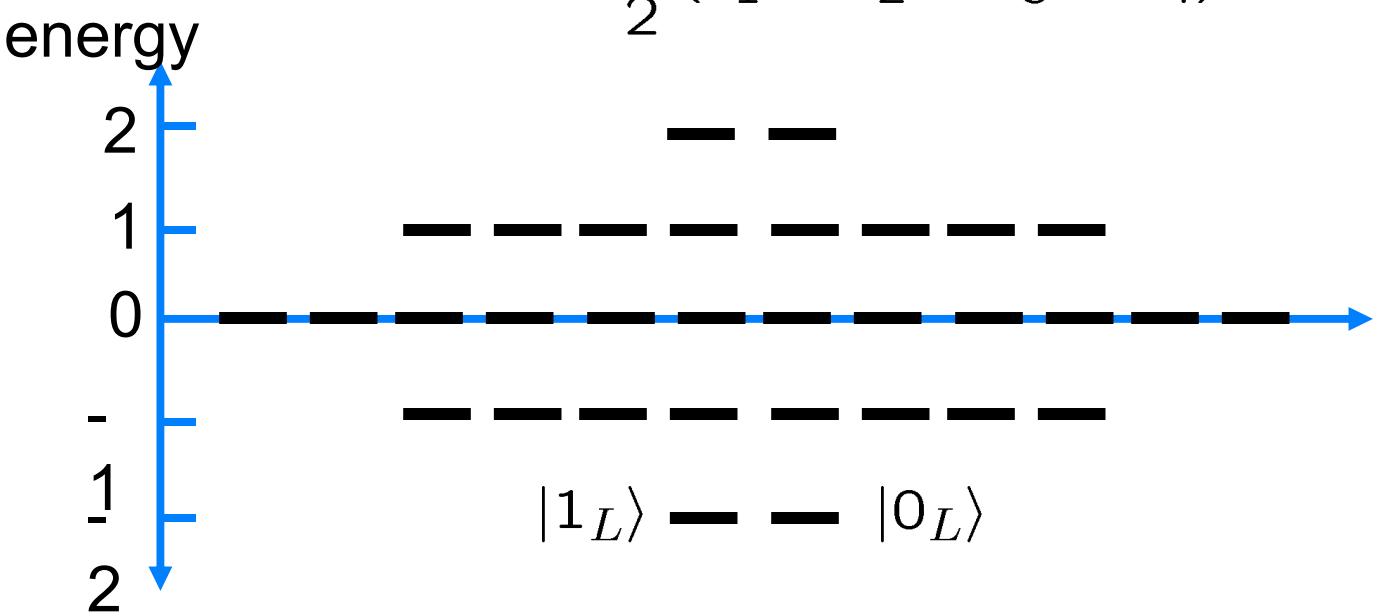
A Start

5 qubit quantum error correcting code

$$s_1 = X \otimes Z \otimes Z \otimes X \otimes I$$
$$s_2 = I \otimes X \otimes Z \otimes Z \otimes X$$
$$s_3 = X \otimes I \otimes X \otimes Z \otimes Z$$
$$s_4 = Z \otimes X \otimes I \otimes X \otimes Z$$

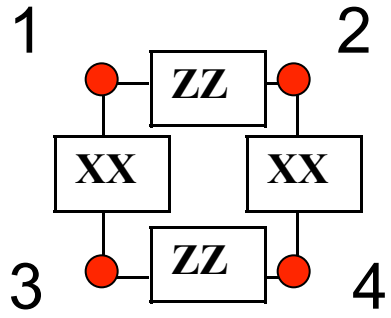
Hamiltonian such that ground state is this error correcting code (ground state is twofold degenerate):

$$H = -\frac{\Delta}{2}(s_1 + s_2 + s_3 + s_4)$$



Two Body?

Problem: four qubit interactions? Bleh.



quantum

Hamiltonian

$$H = -(Z_1Z_2 + Z_3Z_4 + X_1X_3 + X_2X_4)$$

quantum error detecting code

$$S_1 = Z_1Z_2Z_3Z_4$$

$$S_2 = X_1X_2X_3X_4$$

$$\bar{Z}_1 = Z_1Z_2 \qquad \bar{X}_1 = Z_1Z_3$$

$$\bar{Z}_2 = Z_1Z_3 \qquad \bar{X}_2 = X_1X_2$$

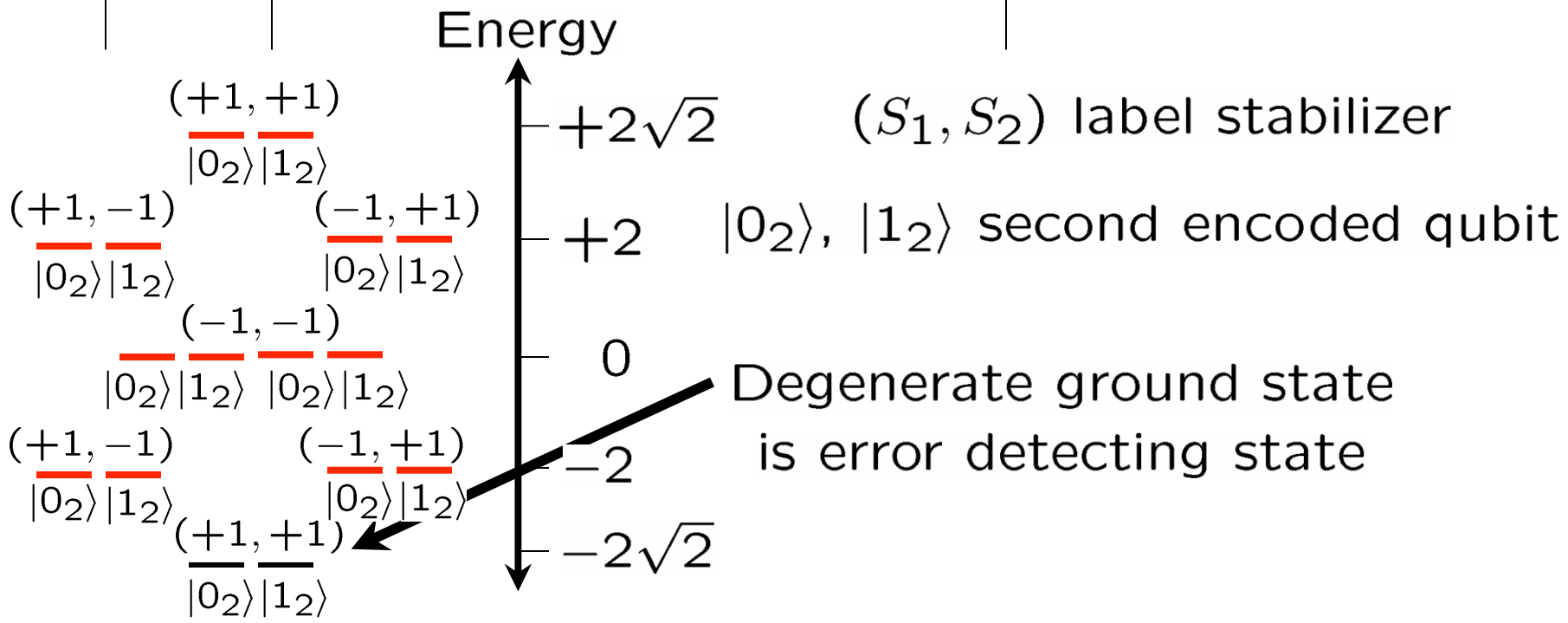
symmetry

$$[H, S_1] = [H, S_2] = 0$$

$$H = -\bar{Z}_1(I + S_1) - \bar{X}_1(I + S_2)$$

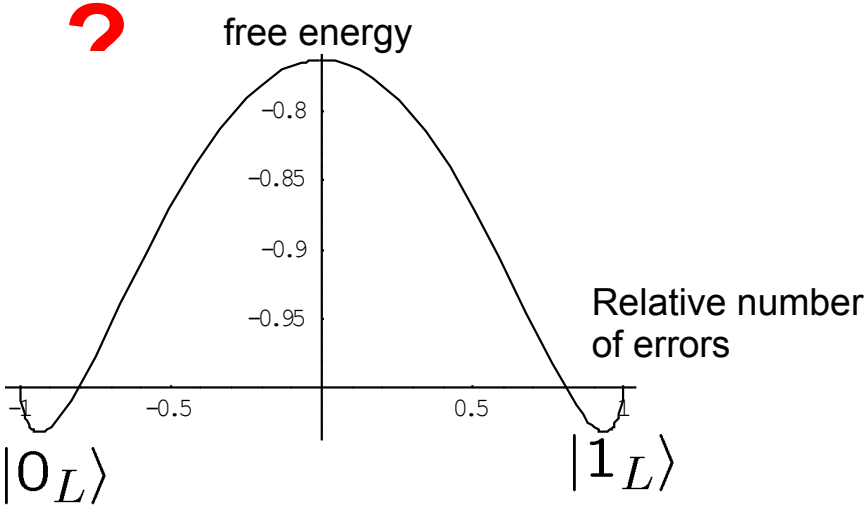
H does not depend on second qubit (use to store encoded qubit)

S_1	S_2	$H = \bar{X}_1(1+S_1) + \bar{Z}_1(1+S_2)$	Eigenvalues
$+1$	$+1$	$2(\bar{X}_1 + \bar{Z}_1)$	$\pm 2\sqrt{2}$
$+1$	-1	$2\bar{X}_1$	± 2
-1	$+1$	$2\bar{Z}_1$	± 2
-1	-1	0	0



ground state is +1 eigenvalue of four qubit interactions

Problem 2



Suppose we want ground state to be a quantum error correcting code which corrects more than two errors using Hamiltonian with two qubit interactions.

$$H = \sum_{ij} \sum_{\alpha,\beta} J_{ij,\alpha\beta} \sigma_\alpha^{(i)} \sigma_\beta^{(j)}$$

These two qubit interactions should be correctible errors: i.e. they take the codespace to outside of the codespace!

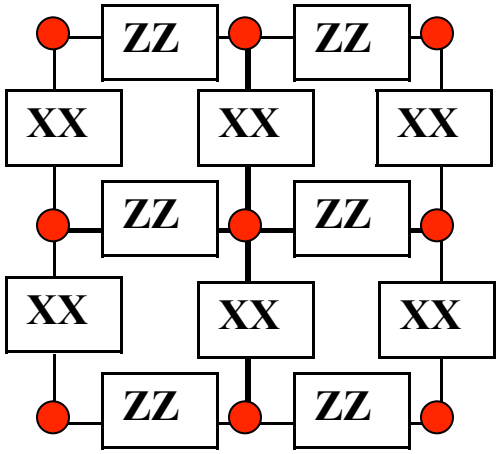
Degenerate Codes to the Rescue

$$s_1 = X \otimes Z \otimes Z \otimes X \otimes I$$
$$s_2 = I \otimes X \otimes Z \otimes Z \otimes X$$
$$s_3 = X \otimes I \otimes X \otimes Z \otimes Z$$
$$s_4 = Z \otimes X \otimes I \otimes X \otimes Z$$

$$H = -\frac{\Delta}{2}(s_1 + s_2 + s_3 + s_4)$$

Some errors, have no effect on the codespace, for example *si*

This leads to the concept of *degenerate* quantum codes



$$H = -J_X \sum_{ij} X_{i,j} X_{i+1,j} - J_Z \sum_{ij} Z_{i,j} Z_{i,j+1}$$

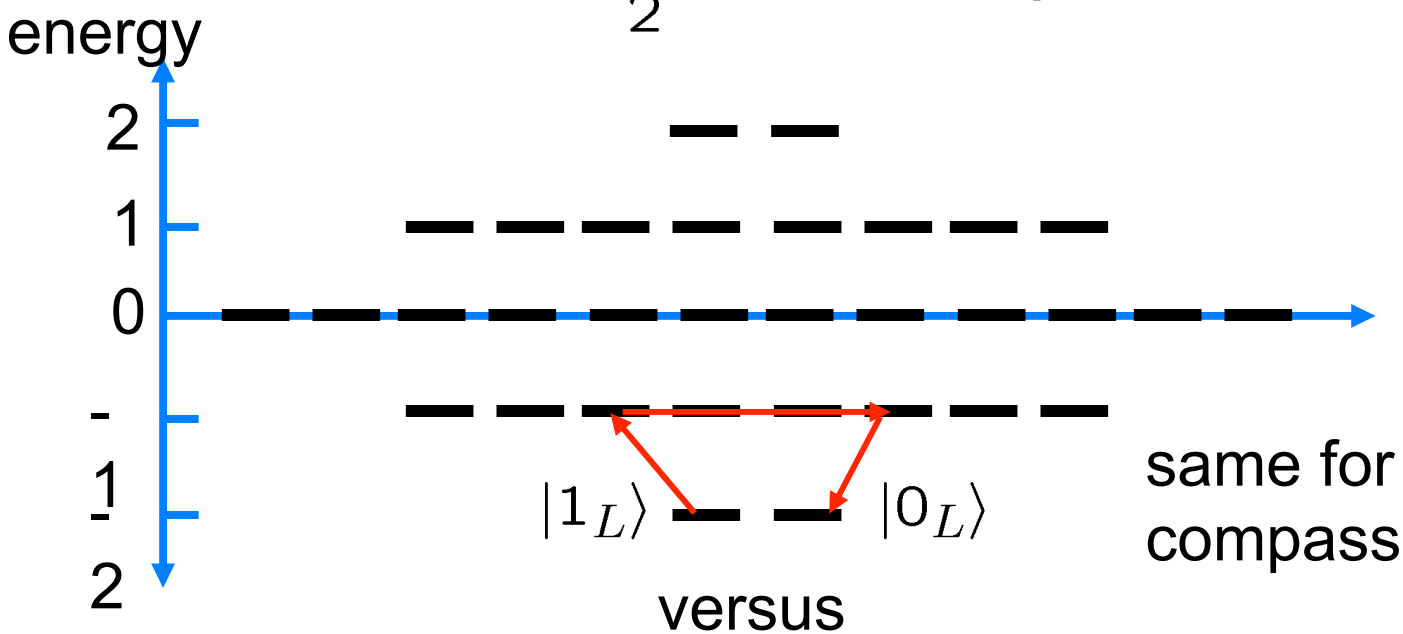
quantum compass model

Ground state is a quantum error correcting code which tolerates ~L/2 errors

[Bacon, Phys. Rev. A, 73, 012340 (2006)]

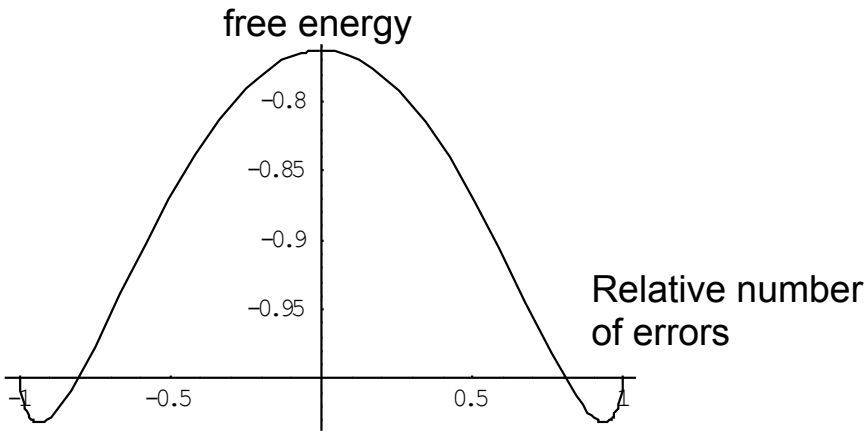
Mountains, Mol

$$H = -\frac{\Delta}{2}(s_1 + s_2 + s_3 + s_4)$$



same for quantum compass model Λ

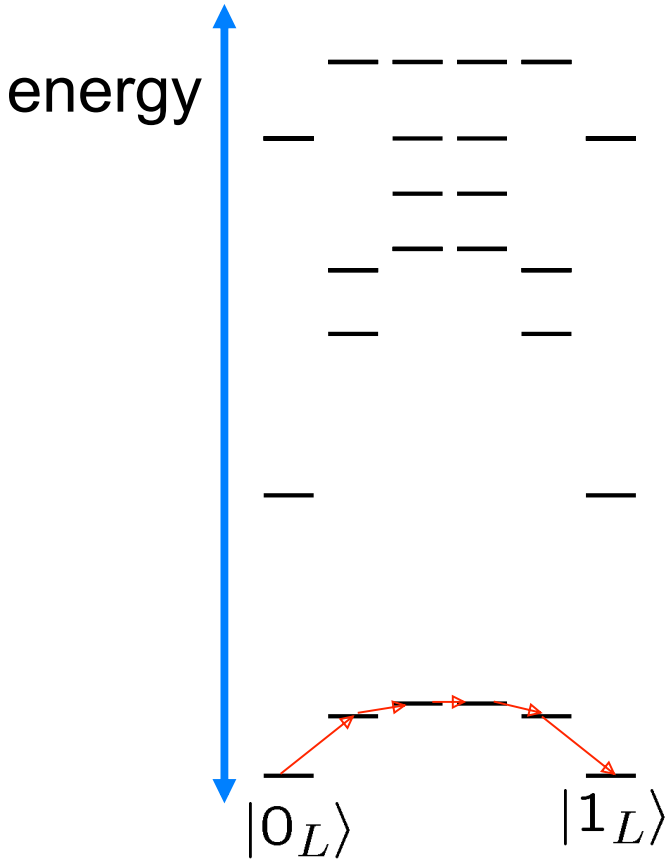
versus



Mountains into Molehills

Long range quantum compass model

$$H = -\sum_{i,j=1}^{n}\sum_{k=1}^{n-1} J_X(k) X_{i,j} X_{i+k,j} + J_Z(k) Z_{i,j} Z_{i,j+k}$$



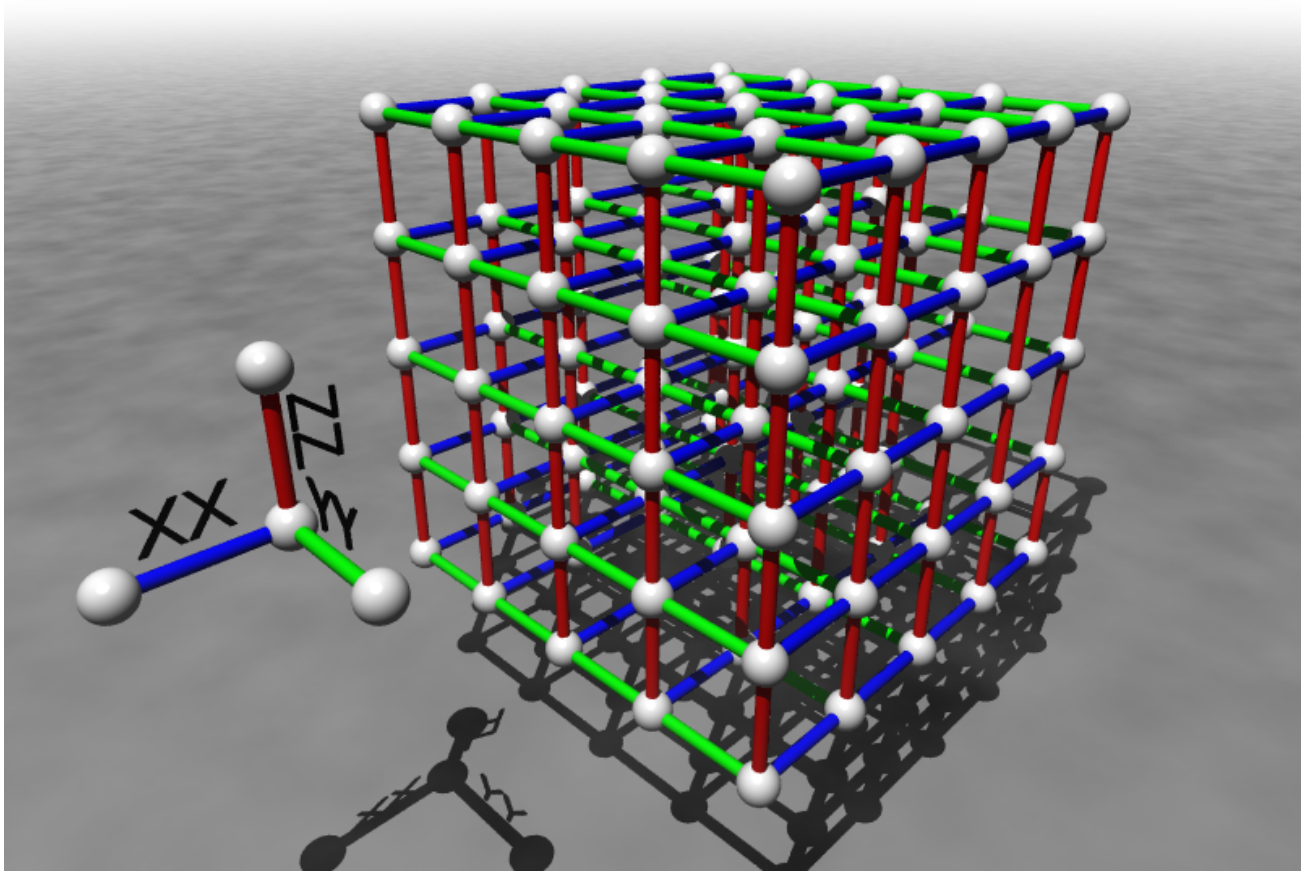
Low energy spectrum for L=4, uniform coupling strength

"Self-correcting"

Bacon 06, in preparation

Into The Third Dimension

$$H = -J_X \sum_{ijk} X_{i,j,k} X_{i+1,j,k} - J_Y \sum_{ijk} Y_{i,j,k} Y_{i,j+1,k} - J_Z \sum_{ijk} Z_{i,j,k} Z_{i,j,k+1}$$



Towards Robust Qubits

Wire many faulty qubits together into a fault-tolerant scheme



Engineer physical systems whose physics enacts error correction

